

Causal Mimicry in Base LLMs

Direction, Dose, and Representation–Output Alignment

Research question. Do base LLMs encode structured causal-relational representations — including causal direction and cause intensity — that are distinct from lexical co-occurrence, and when do these representations control model outputs?

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Key claim. This pilot does not claim that base LLMs perform Pearlian causal inference. It studies **causal mimicry** — the text-learned causal-relational structure that supports causally competent behaviour — and asks a sharper, mechanistic question: **when models encode causal direction, relation type, and cause intensity in their hidden states, and when that structure actually controls their outputs.** The central preliminary result is a **representation–output dissociation**: internal causal-relational structure can be strong while output behaviour is still captured by associative plausibility.

What we mean by “dose.” Each item is a directed causal statement of the form “*X causes Y*” (e.g. *smoking* → *lung cancer*). We vary the **intensity of the cause** with natural-language modifiers rather than the relation itself: *light smoking* → *moderate smoking* → *heavy smoking*, versus the bare *smoking* base. The “dose” question is whether the model’s confidence in the causal claim tracks this graded intensity for *genuine causal* relations while staying flat for anti-causal and spurious ones — a graded analogue of a dose–response curve, read from the model’s own continuation likelihood.

Motivation and core question

Large language models produce fluent causal claims, but output-level competence does not imply causal inference. LLMs score well on some causal benchmarks (*Kiciman et al., 2023*) yet remain prone to causal fallacies (*Joshi et al., 2024*), and their performance is strongly shaped by pretraining exposure to causal text (*Feng et al., 2024*). Theoretical results further caution that observationally trained models cannot, in general, recover causal structure in the strong interventional sense (*Roy & Parbhoo, 2026*).

Under this view the model is, in a sense, always parroting; the scientific question is **when parroting successfully mimics causal competence and when it collapses**. A recent survey makes the corresponding methodological call explicit — to probe internal states in order to separate parroting from reasoning (*Wan et al., 2024*). This pilot takes up that call for **base (pre-RLHF) models**, studying the geometry of text-learned causal-relational structure and the conditions under which it is expressed in outputs, building on the linear-representation view of LLM concept geometry (*Park, Choe & Veitch, 2024*).

The closest concurrent work, *Ding & Zhang (2026)*, reports that hidden states of instruction-tuned models encode causal direction far better than their yes/no answers express it — a “knows-but-doesn’t-say” dissociation. The present pilot extends this in four ways: it works on **base models** (isolating the effect from RLHF), adds a graded **cause-intensity (dose)** axis, scores behaviour with **length-normalised continuation likelihood** rather than yes/no, and surfaces a complementary failure mode — not output silence, but output **over-endorsement** of spurious causal-form statements despite strong internal structure.

Why this matters. The goal is to move beyond asking whether LLMs “know causality” and instead explain the conditions under which causal-relational representations are **encoded, used, bypassed, or overridden**. This matters wherever models are trusted to reason about cause and effect — clinical, scientific, and policy settings — because a model that internally separates causal from spurious relations yet still emits fluent, high-confidence but misleading causal claims is precisely the failure that surface evaluations miss. Characterising *when* internal causal structure reaches the output gives a mechanistic handle on that risk: a route toward auditing causal claims at the representational level, and toward interventions that make a model’s stated causality faithful to what it has actually encoded.

Pilot setup

Base LMs only (pre-RLHF where applicable), spanning **Pythia, OLMo, Qwen2.5, Gemma-2 and Mistral** families. Relations are labelled **causal**, **anti-causal** (reversed), and **spurious / fork** (common-cause, no direct edge). Each item also carries graded cause-intensity prompts (base / low / moderate / high).

- **Behaviour:** length-normalised continuation likelihood and p(correct) over forward / reverse / null continuations.
- **Controls:** modifier-frequency residualisation and a sentence-naturalness (prompt-fluency) control.

- **Representation:** per-layer linear probes on hidden states, leakage-controlled by relation ID (GroupKFold) and benchmarked against TF-IDF and random-label baselines.
- **Direction:** a matched fork-reversal word-order baseline that direction decoding must beat.
- **Alignment & interventions:** probe–output alignment, plus activation steering and cause-token patching used only as intervention diagnostics, not as headline results.

Preliminary findings

Finding 1. Behavioural causal-dose mimicry survives lexical controls.

Across Gemma-2-9B, Mistral-Nemo-12B, OLMo-7B and Qwen2.5-32B, graded cause-intensity prompts produce relation-specific shifts in $p(\text{correct})$. The diagnostic is not strict monotonicity from base to high — the unmodified base statement (“smoking causes cancer”) is already a strong generic causal claim, so low-intensity modifiers often lower confidence. The informative signal is whether confidence **selectively recovers or rises under high-intensity modifiers for genuine causal relations**, while anti-causal and fork/spurious relations show no comparable high-dose residual. **Gemma-2-9B and Qwen2.5-32B give the cleanest behavioural-control case so far:** after residualising $p(\text{correct})$ against modifier frequency, causal residual confidence rises toward high-intensity prompts, anti-causal relations remain negative, and spurious relations return close to zero. Together with the naturalness and co-occurrence controls, this indicates that the dose signature is not reducible to modifier frequency, prompt fluency, or simple lexical co-occurrence.

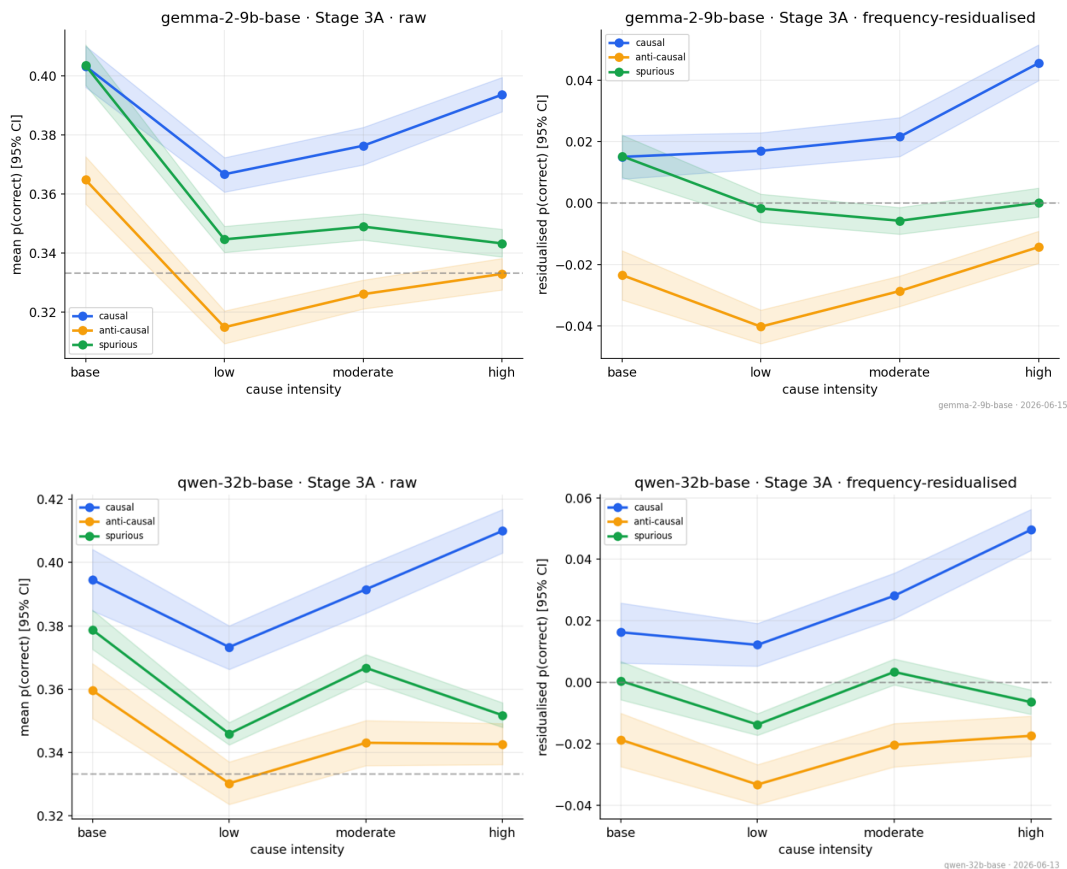


Figure 1. Behavioural cause-intensity effects in Gemma-2-9B and Qwen2.5-32B. Left panels show raw $p(\text{correct})$; right panels show $p(\text{correct})$ residualised against modifier frequency. In both models, causal residual confidence rises toward high-intensity prompts, anti-causal relations remain negative, and spurious relations return close to zero. This indicates that the graded causal-dose signature is not explained by modifier frequency. Bands are 95% group-bootstrap CIs over relation IDs; dashed line marks chance.

Finding 2. The dose signature is not explained by sentence naturalness.

As a frame-fluency control, full-sentence log-probability predicts almost none of the variance in $p(\text{correct})$: in Mistral-Nemo-12B, $R^2 < 0.01$ for causal (0.007), anti-causal (0.003) and spurious (0.008), with the same weak dependence qualitatively across the other base models. With the frequency residualisation, this indicates that the dose signature is reducible to neither intensity-word

frequency nor prompt fluency. Gemma-2-9B shows the same pattern under alternative MORE-dose framings: frame naturalness varies substantially, but sentence naturalness explains less than 1% of variance in $p(\text{correct})$ across causal, anti-causal, and spurious categories. Figure 2 shows the Mistral-Nemo-12B naturalness control; the corresponding Gemma-2-9B scatter plots are included in the repository.

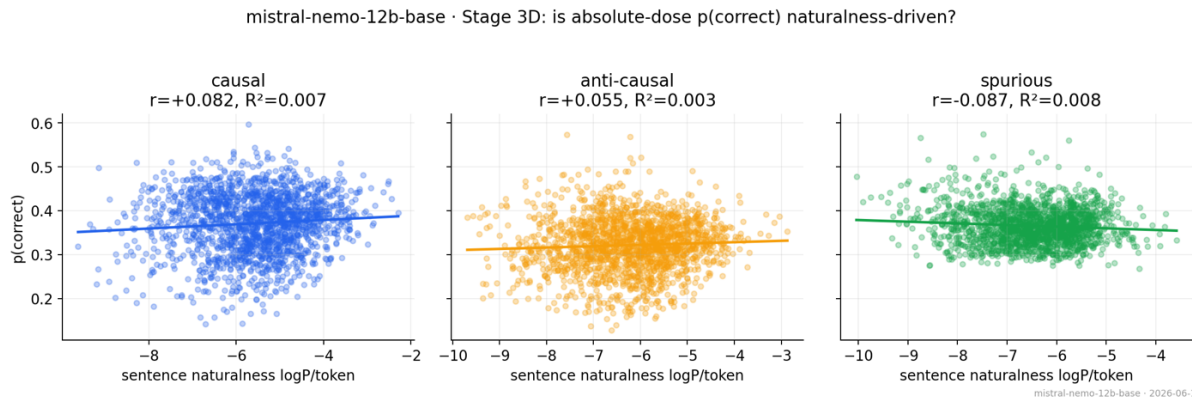


Figure 2. Sentence-naturalness control (Mistral-Nemo-12B, Stage 3D). $p(\text{correct})$ against full-sentence log-probability per token, per category; the fitted slope is near-flat and $R^2 < 0.01$ for causal, anti-causal and spurious. The dose signature is therefore not a frame-fluency artifact.

Finding 3. Hidden states encode relation type and direction beyond lexical baselines.

Some binary contrasts are lexically easy — TF-IDF already separates causal vs spurious — so that contrast is not claimed as evidence of representation. The stronger result is that TF-IDF **fails on the causal-vs-anti direction task and stays near chance on 3-way classification**, whereas hidden-state probes are well above both lexical and random baselines. The model’s hidden states therefore carry direction and joint-category structure not recoverable from surface lexical overlap.

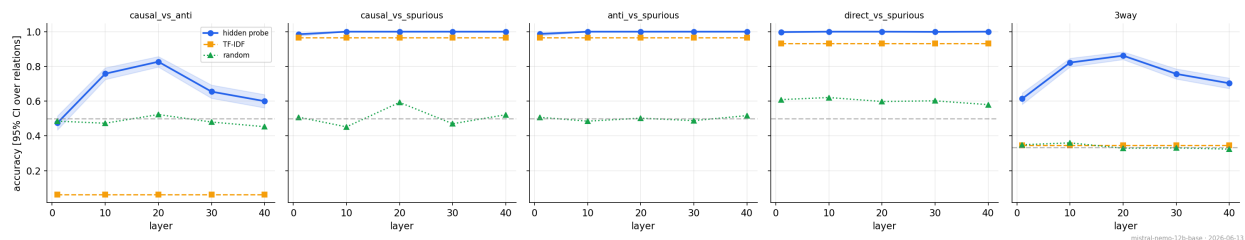


Figure 3. Hidden-state separability vs lexical and random baselines (Mistral-Nemo-12B), per layer. Hidden probes beat TF-IDF and random on causal-vs-anti direction and on 3-way classification; TF-IDF is competitive only on the lexically easy causal-vs-spurious contrast. Accuracy with 95% group-bootstrap CIs over relation IDs.

Finding 4. Causal-direction encoding differs sharply by model family.

In Pythia and Qwen, causal-vs-anti direction is linearly decodable but comparable to or weaker than the fork-reversal word-order baseline, indicating order entanglement. **OLMo-2-13B shows a modest/borderline advantage over the word-order baseline. By contrast, Gemma-2-9B and the Mistral family show much clearer direction-specific structure:** Gemma-2-9B reaches ≈ 0.90 causal-vs-anti accuracy versus ≈ 0.71 for the fork-reversal control, while Mistral-Small-24B reaches ≈ 0.91 versus ≈ 0.75 . Causal direction is therefore **not implemented uniformly across base LMs and is not a simple function of scale**; it varies substantially by model family.

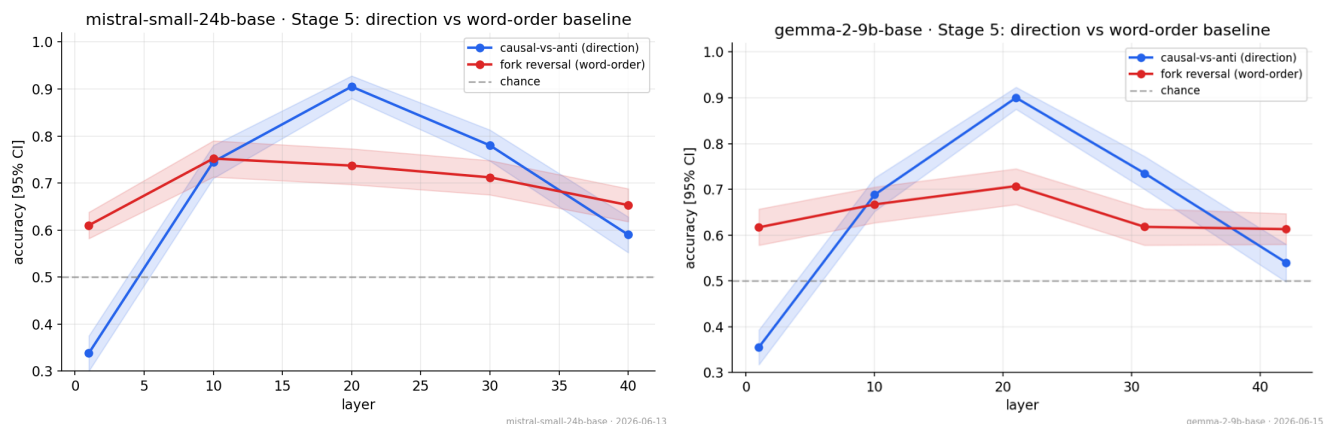


Figure 4. Causal-direction decoding vs a matched word-order baseline Mistral-Small-24B and Gemma-2-9B: direction (blue) peaks ≈ 0.91 in mid-layers, well above the fork-reversal control (red, ≈ 0.75). Gemma-2-9B shows the same pattern even more cleanly (≈ 0.90 vs ≈ 0.71 ; Finding 4, Table 1), whereas Pythia and Qwen sit closer to the word-order floor. 95% group-bootstrap CIs.

Finding 5. Internal causal structure does not always control outputs.

Mistral-Small-24B remains the clearest **representation–output dissociation**. Behaviourally it over-endorses fork/spurious causal-form completions (poor output-level calibration), yet internally it shows strong causal/spurious separability, direction beyond the word-order baseline, and highly structured high-minus-base dose-shift geometry (held-out AUC ≈ 1.0 across mid-layers). Gemma-2-9B provides a complementary case: it shows clean behavioural dose controls, strong causal-direction decoding beyond word order, and high-AUC causal-vs-spurious dose-shift geometry, but a simple global dose-steering vector is **anti-aligned with causal correctness**. The failure mode is therefore not absence of causal-relational structure, but unreliable alignment between internal structure, intervention directions, and output logits.

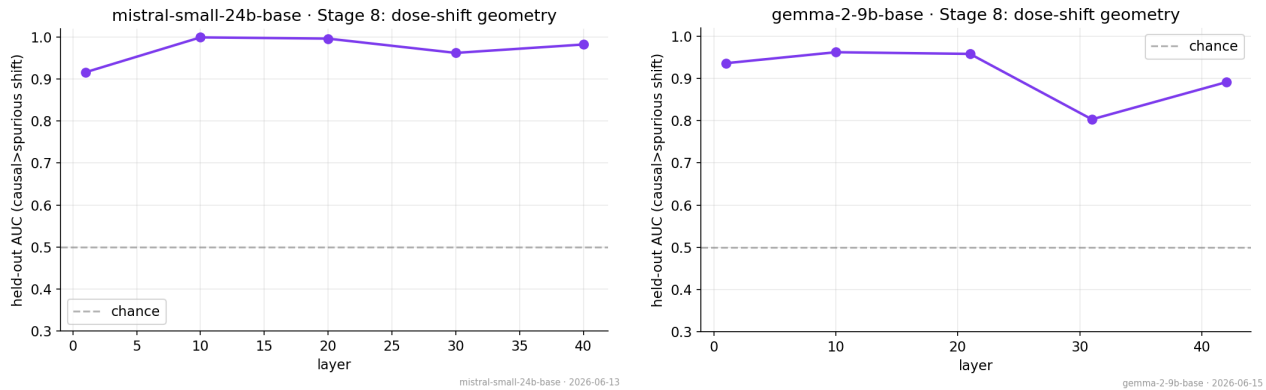


Figure 5. Dose-shift geometry in Mistral-Small-24B (left) and Gemma-2-9B (right). A direction learned on a training split of high-minus-base hidden shifts separates held-out causal from spurious relations with high AUC across layers in both models. This suggests cause intensity induces relation-conditioned hidden-state geometry across multiple base-model families, not only the Mistral family. GroupKFold by relation ID; dashed line marks chance.

Finding 6. Intervention diagnostics suggest distributed, not token-local, dose signals.

As preliminary diagnostics, activation steering suggests that dose-related hidden directions can be behaviourally active in some models, but **not as a simple universal causal-dose knob**. In Gemma-2-9B, for example, strong dose-shift geometry coexists with anti-aligned global dose steering for causal correctness, while cause-token patching does not isolate a clean token-local mechanism. These results are consistent with distributed, relation-conditioned dose representations and are reported as diagnostics, following activation-patching best practice (Zhang & Nanda, 2024), not as headline evidence.

Model family	Direction vs word-order	Relation-type / dose structure	Representation–output picture
Pythia / Qwen	$\approx$ or $<$ baseline	Present	Order-entangled direction
OLMo-2-13B	modest / borderline $>$ baseline	Present	Partial direction beyond order
Gemma-2-9B	0.90 vs ≈ 0.71	Strong behavioural controls + dose geometry	Direction beyond surface order; simple global steering not sufficient
Mistral-Nemo-12B	Exceeds baseline in mid-layers	Strong	Direction beyond surface order
Mistral-Small-24B	0.91 vs ≈ 0.75	Strong + dose geometry	Dissociation: structure not reliably used

Table 1. Model-family contrast: causal-direction decoding vs the word-order baseline, and the representation–output picture.

Working hypothesis and proposed direction

This pilot suggests that causal mimicry in base LMs depends on two separable components: **internal relation geometry** and **output alignment**. Internal relation geometry refers to whether relation type, causal direction, and cause intensity are encoded in hidden states. Output alignment refers to whether this internal structure is actually used to suppress spurious causal-form completions and select the correct continuation.

The central failure mode identified so far is therefore not simply missing causal-relational structure, but representation–output miscalibration. In several models, hidden states contain separable information about causal, anti-causal, and spurious relations, yet this structure is only weakly aligned with output probabilities and can fail to prevent high-confidence spurious causal completions.

The contrast between Gemma-2-9B, Qwen, OLMo and the Mistral family suggests that this is a model-family-dependent routing problem rather than a simple scale effect: some models encode causal direction beyond surface order, while others remain more order-entangled.

The proposed PhD will investigate **when latent causal-relational representations become behaviourally usable**. A working hypothesis is that miscalibration arises from a **late-stage readout bottleneck**: causal-relational geometry may be formed in intermediate hidden states, while final-layer computation and logit readout can still overweight associative plausibility, lexical naturalness, or surface causal-form priors. This hypothesis will be tested rather than assumed, using targeted interventions that distinguish attention-mediated surface-form routing from MLP-mediated relational encoding.

Next steps

- Audit and expand the causal/spurious sets to control for hidden indirect causality.
- Run cleaner same-family scaling ladders to separate scale from family effects.
- Add polarity-dissociation tests — for protective causes, higher intensity should reverse the expected effect (more cause → less effect), separating causal dose from intensity-word salience.
- Strengthen interventions to distinguish distributed dose geometry from token-local mechanisms; extend later to multimodal causal-relational structure if appropriate.

Long-term PhD direction

This pilot would form the first project in a broader PhD programme on when causal-relational structure in foundation models becomes behaviourally usable. The thesis would develop along three connected directions. **First**, I would turn the current text-only causal-mimicry setting into a systematic same-family and cross-family scaling study, testing when relation type, causal direction, cause intensity, and probe-output alignment emerge across base and instruction-tuned models, and why families such as Gemma, Mistral, Qwen and OLMo differ in direction encoding and output alignment. **Second**, I would move from representational diagnostics to mechanistic localisation, using attribution/path patching, causal tracing, and component-level interventions to test whether causal-relational geometry is weakened or bypassed during late-stage readout. If the text-only mechanism is validated, **a later extension** would test whether the same framework transfers to multimodal models, where causal relations can be grounded in visual events, object interactions, counterfactual scenes, and intervention-sensitive physical reasoning. The long-term aim is to explain not merely whether models encode causal structure, **but when such structure is routed into outputs or actions rather than overridden by associative plausibility**.

Feasibility and risks

The project is designed to remain informative even if the current working hypothesis is partially wrong. If late-stage readout is not the main bottleneck, the same experimental framework can test alternative failure modes: diffuse relation representations, lexical confounding, dataset hidden causality, instruction-tuning effects, or non-linear readout mechanisms. The compute requirements are staged: the first phase uses open-weight base models from 410M to 32B with lightweight probing and likelihood scoring; **larger same-family ladders, cross-family comparisons, and multimodal extensions** would be prioritised only after the text-only mechanism is validated.

References

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